

Alex Mercer

OPEN SOURCE SYSTEMS ENGINEER & TECHNICAL EDUCATOR

Email: alex.mercer.dev@gmail.com	GitHub: github.com/alexmercer-dev	Location: San Francisco, CA / Remote
Portfolio: alexmercer.dev	LinkedIn: linkedin.com/in/alexmercer-dev	OSSF Member: Open Source Security Foundation

Versatile **Open Source Systems Engineer, Debugger, Web Developer, and Technical Educator** with over 8 years of experience building and scaling open-source infrastructure, diagnosing complex low-level performance bottlenecks, and creating full-stack web platforms. Active open-source maintainer with over 12,000 combined GitHub stars across infrastructure, kernel diagnostic tools, and developer education frameworks. Dedicated mentor and author who has authored interactive curricula adopted by university CS programs and mentored 20+ Google Summer of Code (GSoC) & Outreachy contributors.

TECHNICAL EXPERTISE

System Administration	Linux Systems Architecture (RHEL/Debian/Arch), Kubernetes, Docker, Ansible, Terraform, Nginx, BGP/Networking, CI/CD Pipelines, Systemd, Prometheus, Grafana, Security Hardening.
Debugging & Diagnostics	eBPF / BCC, GDB, Valgrind, perf, Strace, Wireshark, Core Dump Analysis, Flame Graphs, Memory Leak Tracing, Kernel Panics, Network Latency Profiling.
Web Development	TypeScript, Node.js, Python (FastAPI / Django), Go, React, Next.js, PostgreSQL, Redis, REST & GraphQL APIs, WebSockets, WebAssembly (Wasm), Web Vitals Optimization.
Education & Community	Technical Authoring, Interactive Workshops, GSoC/Outreachy Mentorship, Open Source Governance, API Documentation, Public Speaking, Hands-on Lab Design.

KEY OPEN SOURCE CONTRIBUTIONS & PROJECTS

Creator & Primary Maintainer — <i>KernelTrace-CLI</i>	github.com/alexmercer-dev/kerneltrace-cli (4.8k ★)
- Engineered an eBPF-powered real-time system call tracing and performance diagnostic utility written in Go and Rust for Linux servers. - Enabled enterprise sysadmins to identify sub-millisecond I/O bottlenecks and memory leaks with minimal CPU overhead (<0.5%). - Created comprehensive interactive CLI tutorials, enabling 150+ community contributors to submit kernel telemetry probes.	
Core Contributor & Web Architect — <i>OpenCloud Dashboard</i>	github.com/opencloud/dashboard (7.2k ★)
- Architected high-performance web UI in React, TypeScript, and WebSockets for real-time monitoring of open-source container clusters. - Implemented custom interactive log-viewer with WebGL acceleration capable of streaming 50,000 log events/sec without browser lag. - Authored full documentation suite and developer onboarding guides, reducing pull request review cycles by 35%.	

PROFESSIONAL EXPERIENCE

Senior Systems Administrator & Open Source Lead — <i>FossSphere Infrastructure</i>	2022 – Present
- Managed multi-region hybrid Linux cluster (600+ bare-metal and cloud nodes) powering open-source mirror networks with 99.99% uptime. - Diagnosed and patched critical concurrency and kernel-level socket memory leaks in high-throughput Nginx proxy nodes using eBPF and GDB. - Designed and deployed an open-source web-based metrics platform (FastAPI, React, TimescaleDB) to monitor cluster health and automate triage alerts. - Spearheaded internal technical education series, teaching 40+ cloud engineers advanced Linux system administration, shell scripting, and crash dump analysis.	

Full-Stack Web Developer & Technical Educator — CodeLab Open Learning

2019 – 2022

- Developed open-source web platform for interactive browser-based coding labs, serving over 300,000 monthly active tech learners.
- Built browser-sandboxed terminal emulator using WebAssembly (Wasm) and Node.js, allowing students to execute real Linux commands safely.
- Authored 25+ comprehensive open-access learning modules on web performance, RESTful API security, and systems debugging.
- Mentored 12 Google Summer of Code (GSoC) students in building full-stack web features and open-source DevOps tooling.

Linux Systems Administrator & Systems Debugger — Apex Data Systems

2016 – 2019

- Automated server provisioning, configuration management, and patching across 200+ Debian and RHEL instances using Ansible and Bash.
- Performed root-cause analysis (RCA) on production system crashes, network packet loss, and database lockups using `perf`, `strace`, and GDB.
- Created and hosted bi-weekly internal Linux shell bootcamps and debugging workshops for junior developers and sysadmin staff.

EDUCATION, CERTIFICATIONS & COMMUNITY

Education & Certifications

- **B.S. in Computer Science** — Univ. of California (2012–2016)
- **Linux Foundation Certified System Administrator (LFCS)**
- **Certified Kubernetes Administrator (CKA)**
- **Red Hat Certified Engineer (RHCE)**

Talks & Mentorship

- **Speaker, Open Source Summit:** *"Debugging Linux Systems in Production with eBPF"*
- **GSoC / Outreachy Org Admin & Mentor:** 2020 – Present
- **Host, FOSS Developer Podcast & Video Tutorials:** 45k+ Subscribers